Data Analyst in Tableau with AI

A basic explanation of the six core topics

#	Topic	Focus
1	Data Foundations	Excel, SQL basics, data cleaning habits
2	Tableau Public Basics	Connecting data, dimensions, measures, first charts
3	Calculations	Calculated fields, LOD expressions, table calculations
4	Dashboard Design	Layout, filters, actions, colour and accessibility
5	Advanced Visuals	Maps, dual axis, parameters, forecasting
6	AI Assisted Insights and Portfolio	Prompting AI for analysis, publishing to Tableau Public

1. Data Foundations

Good dashboards start with good data. Before opening Tableau, learn to handle data in Excel and SQL.

Excel essentials

- **Tables and structure:** one row per record, one column per field, a single header row, no merged cells.
- **Core functions:** SUM, AVERAGE, COUNTIF, IF, XLOOKUP (or VLOOKUP) to bring in related data.
- **PivotTables:** quickly summarise data by category, the same idea Tableau uses visually.
- **Save as CSV or XLSX:** both connect easily to Tableau.

SQL basics

SQL is the language used to ask questions of databases. Five clauses cover most beginner needs:

```
SELECT region, SUM(sales) AS total_sales
FROM orders
WHERE order_date >= '2025-01-01'
GROUP BY region
ORDER BY total_sales DESC;
```

Also learn **JOIN** (combine tables using a shared key such as customer_id) and aggregates like COUNT, AVG, MIN, MAX.

Data cleaning habits

- Keep the raw file untouched; clean a copy.
- Remove duplicates and fix obvious typos (for example 'Delhi' vs 'delhi ').
- Handle missing values deliberately: fill, flag or exclude, and note which.
- Use consistent data types: dates as dates, numbers as numbers.
- Standardise names, units and currencies.
- Document every change so your work can be repeated.

> **Practice task**
> Pick a free dataset (for example sales or sports data), find three data quality problems and fix them in Excel.

2. Tableau Public Basics

Tableau Public is a free version of Tableau for building and sharing visualisations online.

Connecting data

Open Tableau Public and choose a connector on the Connect pane. Excel, text/CSV and Google Sheets are the most common. Drag tables onto the canvas to relate or join them, then open a sheet to start building.

Dimensions and measures

	Dimensions	Measures
Meaning	Categories used to slice data	Numbers that can be aggregated
Examples	Region, Product, Date, Customer	Sales, Profit, Quantity
Appears as	Blue pills (discrete) or green (continuous dates)	Green pills, aggregated as SUM, AVG and so on
Role	Define the level of detail and groupings	Define the size or height of marks

Discrete (blue) vs continuous (green)

Blue pills create separate headers or labels. Green pills create a continuous axis. This choice changes how a chart looks, so check it whenever a chart behaves unexpectedly.

Your first charts

Chart	Use it to	How
Bar chart	Compare categories	Dimension on Rows, measure on Columns
Line chart	Show change over time	Date on Columns, measure on Rows
Scatter plot	See relationship between two measures	One measure on each axis, dimension on Detail
Text table	Show exact numbers	Dimensions on Rows, measures in Text

Key tools: the **Marks card** (Colour, Size, Label, Detail, Tooltip), **Filters shelf**, and **Show Me** for suggested chart types.

> **Practice task**
> Build a bar chart of Sales by Category, then a line chart of monthly Sales, and add colour for Profit.

3. Calculations

Calculations let you create new information that does not exist in your raw data.

Calculated fields

Create one with Analysis > Create Calculated Field. They work row by row or on aggregates.

```
// Profit ratio
SUM([Profit]) / SUM([Sales])

// Category flag
IF [Sales] > 1000 THEN 'High' ELSE 'Low' END

// Days to ship
DATEDIFF('day', [Order Date], [Ship Date])
```

LOD (Level of Detail) expressions

LODs compute values at a level you choose, independent of what is in the view.

Type	Meaning	Example
FIXED	Calculate at the dimensions you name	{ FIXED [Customer] : SUM([Sales]) }
INCLUDE	Add a dimension to the view's level	{ INCLUDE [Customer] : SUM([Sales]) }
EXCLUDE	Remove a dimension from the view's level	{ EXCLUDE [Region] : SUM([Sales]) }

Common use: total spend per customer, then average it across all customers.

Table calculations

These run on the results already shown in the view, after filters. Right-click a measure pill and choose Quick Table Calculation.

- **Running total:** cumulative sales over time.
- **Percent of total:** each bar's share of the whole.
- **Rank:** position of each item.
- **Moving average:** smooths noisy trends, for example WINDOW_AVG(SUM([Sales]), -2, 0).

Remember: use *Compute using* to control the direction (table across, down, or specific dimension) a table calculation follows.

> **Practice task**
> Create a profit ratio field, a FIXED LOD for sales per customer, and a running total of sales by month.

4. Dashboard Design

A dashboard combines several sheets to tell one clear story for a specific audience.

Layout

- Start with the question the audience needs answered and put the most important view top-left.
- Use **tiled** layout for tidy alignment and **floating** only for things like legends or logos.
- Use horizontal and vertical **containers** to keep spacing consistent.
- Set a fixed size or build a **device layout** for phone and tablet.
- Keep it simple: roughly 3 to 5 views, with a clear title.

Filters

Right-click a filter and choose *Apply to Worksheets* to control which sheets it affects. Show only a few key filters (date, region, category). Too many controls confuse users.

Actions

Action	What it does
Filter	Clicking a mark in one sheet filters other sheets
Highlight	Emphasises related marks across sheets
URL	Opens a web page, for example a source or profile link
Go to Sheet	Navigates between dashboards

Colour and accessibility

- Use colour with purpose: one highlight colour, greys for context.
- Choose colour-blind-friendly palettes; avoid red and green as the only difference.
- Keep strong contrast between text and background.
- Do not rely on colour alone: add labels, shapes or clear legends.
- Use readable font sizes and add alt text and clear titles.
- Add meaningful tooltips and remove clutter such as heavy gridlines.

> **Practice task**
> Combine 3 sheets into one dashboard with a filter action, a highlight action and a colour-blind-safe palette.

5. Advanced Visuals

Once the basics are solid, these techniques make your analysis richer and more interactive.

Maps

Tableau recognises geographic fields such as Country, State and City (shown with a globe icon). Double-click one to create a **symbol map** (circles sized by a measure) or choose **filled map** to colour regions. Check unmatched locations and fix them with Edit Locations.

Dual axis charts

A dual axis combines two measures on one chart, for example Sales as bars and Profit ratio as a line. Right-click the second measure and choose *Dual Axis*, then synchronise axes when units match. Use sparingly; confusing double axes are a common mistake.

Parameters

A parameter is a user-controlled value that is not tied to the data. Create one, show the parameter control, then use it inside a calculated field.

```
// Parameter [Select Measure] = Sales / Profit
CASE [Select Measure]
WHEN 'Sales' THEN SUM([Sales])
WHEN 'Profit' THEN SUM([Profit])
END
```

Typical uses: switch measures, set a threshold, choose a top N.

Forecasting

On a line chart with a continuous date, open the Analytics pane and drag *Forecast* into the view. Tableau uses exponential smoothing to extend the trend and shows a prediction interval. Treat forecasts as estimates; they work best with enough history and clear seasonality. Trend lines and reference lines are also found in the Analytics pane.

> **Practice task**
> Build a map of sales by state, a bar and line dual axis, and a parameter that switches between Sales and Profit.

6. AI Assisted Insights and Portfolio

AI can speed up thinking, but you remain responsible for checking and communicating the results.

Prompting AI for analysis

AI assistants (such as Claude) can help you plan analysis, explain formulas, write SQL, and suggest chart types. A good prompt includes context, data description, task and desired format.

> **Example prompt**
> I am analysing a retail sales dataset with columns: Order Date, Region, Category, Sales, Profit. Suggest 5 questions a manager might ask, the best Tableau chart for each, and the calculated fields needed. Present as a table.

Use AI for	Example
Ideas	Which questions and KPIs matter for this data?
Code help	Write a SQL query for monthly sales by region
Tableau help	Explain the difference between FIXED and INCLUDE with an example
Debugging	Why does my calculated field return Null?
Storytelling	Draft three plain-language takeaways from this chart

Use AI safely

- Always verify numbers and formulas yourself; AI can be confidently wrong.
- Never paste private, confidential or personal data into an AI tool.
- Tableau's own AI features are mainly in paid Tableau products; Tableau Public is best paired with an external assistant.

Publishing to Tableau Public

- Create a free Tableau Public account.
- In Tableau Public choose File > Save to Tableau Public and name the workbook.
- Remember: everything published is **public**. Use only open or sample data.
- Check how the dashboard looks on desktop and mobile, then add a description and tags.
- Share the link on your CV, LinkedIn and GitHub.

Building a portfolio

- Aim for 3 to 5 strong projects across different topics (sales, sports, health, finance).
- For each, state the business question, the data source, your method and the key insight.
- Prefer quality over quantity: clean design, clear story, accurate numbers.

> **Practice task**
> Use an AI assistant to plan one project, build it in Tableau, check every number, publish it, and write a 3-sentence summary.

Quick roadmap

Stage	Goal
Foundations	Clean data confidently with Excel and SQL
Tableau basics	Build bar, line and scatter charts
Calculations	Write fields, LODs and table calcs
Dashboards	Tell a story with filters and actions
Advanced	Maps, dual axis, parameters, forecasts
Portfolio	Publish 3 to 5 projects with AI support